



Lecturer,
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CSE 2201 : Algorithms

Lecture 06: Memory

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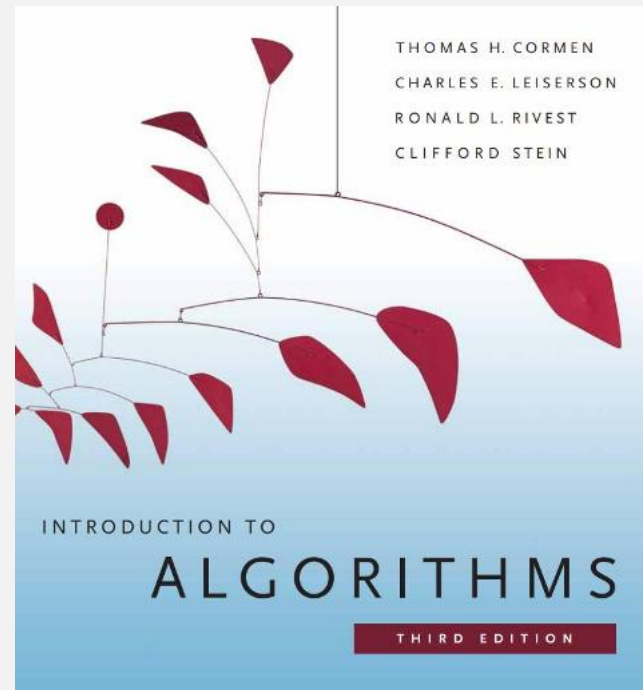


Class Code
GVWTHYJV



Reference Book

- Introduction to Algorithms, 3rd edition. T.H. Cormen, C.E. Leiserson, R.L. Rivest, C. Stein



Distribution of Marks

Assessments	%
Mid Term Examination	25
Final Examination	40
Attendance & class performance	10
Assignment & Quizzes	25
Total	100



PRINCETON
UNIVERSITY

Basics

Bit. 0 or 1.

Byte. 8 bits.

Megabyte (MB). 1 million or 2^{20} bytes.

Gigabyte (GB). 1 billion or 2^{30} bytes.

NIST

most computer scientists



64-bit machine. We assume a 64-bit machine with 8 byte pointers.

- Can address more memory.
- Pointers use more space.



some JVMs "compress" ordinary object pointers to 4 bytes to avoid this cost

Typical memory usage for primitive types and arrays

type	bytes
boolean	1
byte	1
char	2
int	4
float	4
long	8
double	8

for primitive types

type	bytes
char[]	$2N + 24$
int[]	$4N + 24$
double[]	$8N + 24$

for one-dimensional arrays

type	bytes
char[][]	$\sim 2 M N$
int[][]	$\sim 4 M N$
double[][]	$\sim 8 M N$

for two-dimensional arrays

Typical memory usage for objects in Java

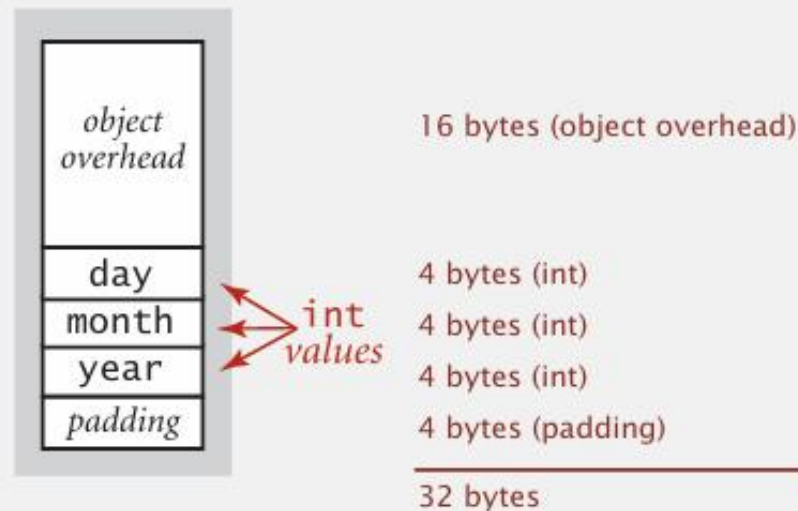
Object overhead. 16 bytes.

Reference. 8 bytes.

Padding. Each object uses a multiple of 8 bytes.

Ex 1. A Date object uses 32 bytes of memory.

```
public class Date
{
    private int day;
    private int month;
    private int year;
    ...
}
```



Typical memory usage for objects in Java

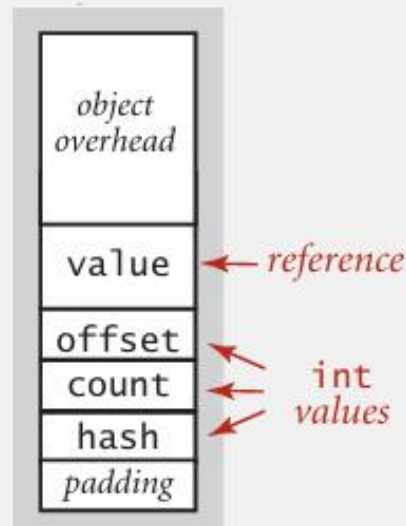
Object overhead. 16 bytes.

Reference. 8 bytes.

Padding. Each object uses a multiple of 8 bytes.

Ex 2. A virgin String of length N uses $\sim 2N$ bytes of memory.

```
public class String
{
    private char[] value;
    private int offset;
    private int count;
    private int hash;
    ...
}
```



16 bytes (object overhead)

8 bytes (reference to array)
2N + 24 bytes (char[] array)

4 bytes (int)

4 bytes (int)

4 bytes (int)

4 bytes (padding)

2N + 64 bytes

Typical memory usage summary

Total memory usage for a data type value:

- Primitive type: 4 bytes for int, 8 bytes for double, ...
- Object reference: 8 bytes.
- Array: 24 bytes + memory for each array entry.
- Object: 16 bytes + memory for each instance variable + 8 bytes if inner class (for pointer to enclosing class).
- Padding: round up to multiple of 8 bytes.

Shallow memory usage: Don't count referenced objects.

Deep memory usage: If array entry or instance variable is a reference, add memory (recursively) for referenced object.

Example

- Q. How much memory does `WeightedQuickUnionUF` use as a function of N ?
Use tilde notation to simplify your answer.

```
public class WeightedQuickUnionUF
{
    private int[] id;
    private int[] sz;
    private int count;

    public WeightedQuickUnionUF(int N)
    {
        id = new int[N];
        sz = new int[N];
        for (int i = 0; i < N; i++) id[i] = i;
        for (int i = 0; i < N; i++) sz[i] = 1;
    }
    ...
}
```

← 16 bytes
(object overhead)

← 8 + (4N + 24) each
reference + int[] array

← 4 bytes (int)

← 4 bytes (padding)

- A. $8N + 88 \sim 8N$ bytes.

Turning the crank: summary

Empirical analysis.

- Execute program to perform experiments.
- Assume power law and formulate a hypothesis for running time.
- Model enables us to **make predictions**.

Mathematical analysis.

- Analyze algorithm to count frequency of operations.
- Use tilde notation to simplify analysis.
- Model enables us to **explain behavior**.



Scientific method.

- Mathematical model is independent of a particular system; applies to machines not yet built.
- Empirical analysis is necessary to validate mathematical models and to make predictions.

Reference

- <http://coursera.org/learn/algorithms-part1/>



Thank You

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